

Scratch Org Creator

User Guide — Step-by-step walkthrough for creating Salesforce scratch orgs

Prerequisites

Before you begin, ensure the following are installed and configured on your machine:

#	Requirement	How to Verify	Install Instructions
1	Salesforce CLI (sf)	Run <code>sf --version</code>	Download or <code>npm install -g @salesforce/cli</code>
2	Python 3	Run <code>python3 --version</code>	Pre-installed on macOS. Windows: python.org
3	Dev Hub Enabled	Setup → Dev Hub → "Enabled"	Setup → Dev Hub → Toggle "Enable Dev Hub" ON
4	Web Browser	Chrome, Safari, Firefox, or Edge	Already available

Files Required

You need **one single file** to get started:

File	Description
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<code>scratch-org-creator.sh</code>	Self-contained installer script (~191 KB). Contains everything — no other files needed.
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Alternatively, if working from the source repository:

File / Folder	Description
<code>create-scratch-org.sh</code>	Main launcher script
<code>scratch-org-ui/app.py</code>	Python Flask backend
<code>scratch-org-ui/features.json</code>	296 Salesforce features database
<code>scratch-org-ui/settings.json</code>	39 categorized org settings
<code>scratch-org-ui/templates/index.html</code>	Web UI interface

Getting Started

Open your terminal and run:

```
bash scratch-org-creator.sh
```

The tool will automatically:

- Verify Python 3 and Salesforce CLI are installed
- Install Flask (first time only, takes a few seconds)
- Start a local web server on port 8484
- Open your default browser to the interactive UI

```
Scratch Org Creator v1.0
```

```
[i] Starting web server on http://localhost:8484 ...  
[✓] Web UI running at: http://localhost:8484
```

Press Ctrl+C to stop the server when done.

Step 1 Select Your Dev Hub

Scratch Org Creator Salesforce DX Interactive Setup

1. Dev Hub 2. Definition 3. Features 4. Settings 5. Create

Dev Hub Connection Required

Select an existing authorized Dev Hub or enter a new one. The Dev Hub org is required to create scratch orgs.

CCIPracticeDevHub s.dutta380@gmail.com

DevHub duttasoumyadeep592.fada5396e403@agentforce.com

Th-DevHub soumyadeep.dutta@health.telstra.com.pbo

Authorize a New Dev Hub

This will open a Salesforce login page in your browser. Log in to the org you want to use as a Dev Hub.

DevHub Production / Developer (login.salesforce.com)

Authorize Dev Hub

Dev Hub Alias or Username

DevHub

Select from the list above, authorize a new one, or type an alias/username directly.

Next: Definition

JSON Preview Live

```
{
  "orgName": "My Scratch Org",
  "edition": "Developer",
  "hasSampleData": true,
  "features": [
    "AnalyticsAdminPerms",
    "DevelopmentWave",
    "CustomerExperienceAnalytics"
  ],
  "settings": {
    "securitySettings": {
      "enableAdminLoginAsAnyUser": true
    },
    "passwordPolicies": {
      "enableSetPasswordInApi": true
    },
    "languageSettings": {
      "enableTranslationWorkbench": true
    }
  }
}
```

Copy JSON **Download**

What is a Dev Hub?

A Dev Hub is the Salesforce org that manages and creates your scratch orgs. You must have one authorized before proceeding.

What to do:

1. **If your Dev Hub is already listed** — Click on it (e.g., "CCIPracticeDevHub", "DevHub", "Th-DevHub"). The alias auto-fills below.
2. **If no Dev Hubs appear** — Use the "Authorize a New Dev Hub" section:
 - Enter an alias (any name you choose)
 - Select login type: "Production / Developer" for most orgs
 - Click **Authorize Dev Hub**
 - A Salesforce login page opens — log in and grant access
 - Once done, your org appears in the list
3. Click **Next: Definition** to continue.

Tip: You can have multiple Dev Hubs authorized. Pick the one appropriate for your project.

Step 2 Configure Org Definition

Scratch Org Creator

Salesforce DX Interactive Setup

1. Dev Hub

2. Definition

3. Features

4. Settings

5. Create

Org Definition

Core Settings

Scratch Org Alias

my-scratch-org2

Org Display Name

My Scratch Org

Edition

Developer

Duration (days)

14

Admin Username (optional)

admin@example.com

Country Code (optional)

US

Description (optional)

Purpose of this scratch org

Language (optional)

en_US

Release (optional)

Default (Current)

Include sample data

Back

Next: Features

JSON Preview

Live

```
{
  "orgName": "My Scratch Org",
  "edition": "Developer",
  "hasSampleData": true,
  "features": [
    "AnalyticsAdminPerms",
    "DevelopmentWave",
    "CustomerExperienceAnalytics"
  ],
  "settings": {
    "securitySettings": {
      "enableAdminLoginAsAnyUser": true
    },
    "passwordPolicies": {
      "enableSetPasswordInApi": true
    },
    "languageSettings": {
      "enableTranslationWorkbench": true
    }
  }
}
```

Copy JSON

Download

This step defines the basic properties of your scratch org.

Field Reference:

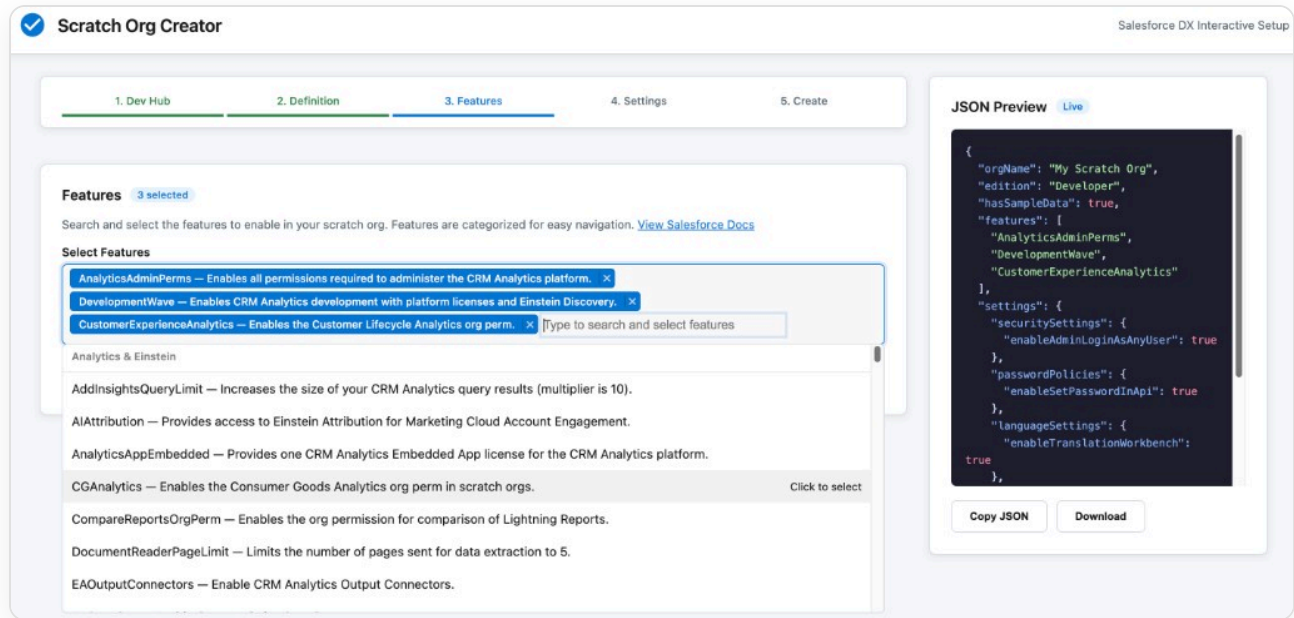
Field	What to Enter	Required?
Scratch Org Alias	Short identifier (e.g., "my-feature-org")	Yes
Org Display Name	Human-readable name (e.g., "Feature Testing")	Yes
Edition	Developer, Enterprise, Group, or Professional	Yes
Duration (days)	1–30 days. Drag the slider to set.	Yes
Admin Username	Leave blank to auto-generate, or enter a valid email format	No
Country Code	2-letter ISO code (US, IN, AU, etc.)	No

Description	Brief purpose of the org	No
Language	Language code (e.g., "en_US")	No
Release	Default (Current), Previous, or Preview	No
Include sample data	Toggle ON to pre-populate sample records	No

Tip: Use "Developer" edition for standard development. Choose "Enterprise" only when you need to test Enterprise-specific features. A duration of 7–14 days is typical.

Click **Next: Features** to continue.

Step 3 Select Features



What are Features?

Features enable specific Salesforce capabilities in your scratch org. There are **296 features** available, organized into categories like Analytics, Sales Cloud, Service Cloud, Marketing, etc.

What to do:

1. **Click the search box** — It says "Type to search and select features"
2. **Search or browse** — Type a keyword (e.g., "Analytics") to filter, or scroll through categories
3. **Click a feature** to select it — it appears as a blue tag in the input
4. **Remove a feature** by clicking the x on its tag

The counter (e.g., "4 selected") updates as you add/remove features. The features list shows the feature name and a brief description for each option.

Features That Require Values

The screenshot shows the 'Scratch Org Creator' application in the 'Features' step. The top navigation bar includes '1. Dev Hub', '2. Definition', '3. Features' (active), '4. Settings', and '5. Create'. The main content area is titled 'Features' with a sub-header '4 selected'. It contains a search bar and a list of four selected features: 'AnalyticsAdminPerms', 'DevelopmentWave', 'CustomerExperienceAnalytics', and 'AddInsightsQueryLimit'. Below the list is a 'Feature Values' section for 'AddInsightsQueryLimit' with an input field showing '1-30' and a range '(1-30)'. At the bottom are 'Back' and 'Next: Settings' buttons. On the right, a 'JSON Preview' panel shows the generated JSON configuration, including the selected features and their values. The JSON is as follows:

```
{
  "orgName": "My Scratch Org",
  "edition": "Developer",
  "hasSampleData": true,
  "features": [
    "AnalyticsAdminPerms",
    "DevelopmentWave",
    "CustomerExperienceAnalytics",
    "AddInsightsQueryLimit"
  ],
  "settings": {
    "securitySettings": {
      "enableAdminLoginAsAnyUser": true
    },
    "passwordPolicies": {
      "enableSetPasswordInApi": true
    },
    "languageSettings": {
      "enableTranslationWorkbench": true
    }
  }
}
```

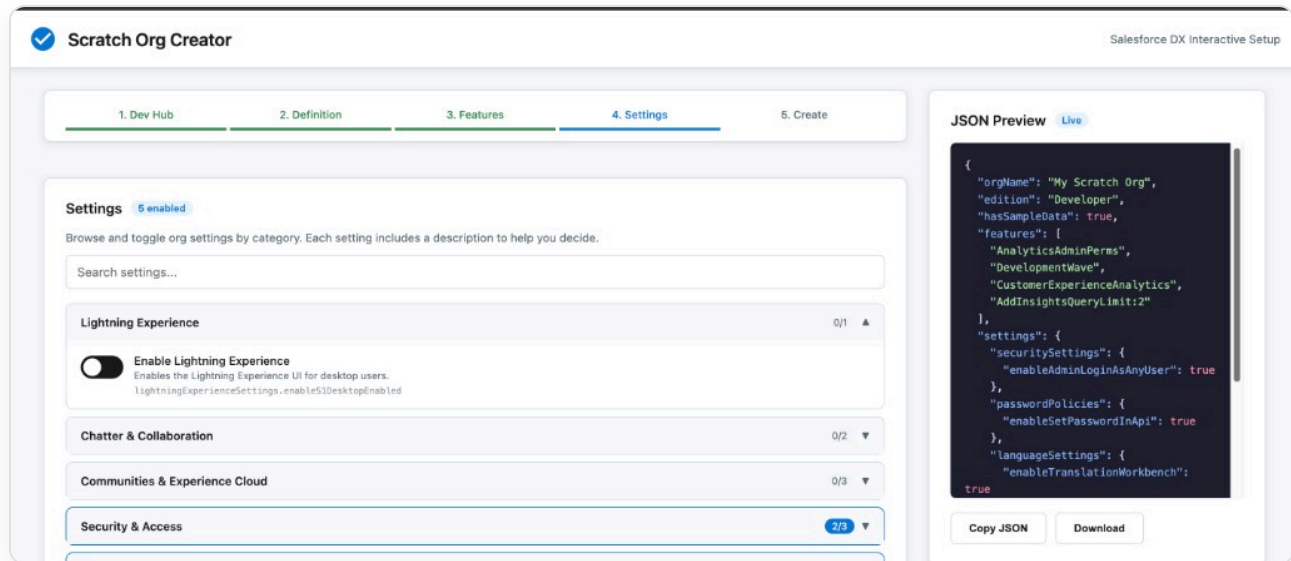
Buttons for 'Copy JSON' and 'Download' are located at the bottom of the JSON preview panel.

Some features need a numeric parameter. When you select one (like **AddInsightsQueryLimit**), a "Feature Values" section appears below with an input field showing the valid range (e.g., 1–30).

Tip: The JSON Preview on the right updates live as you make selections. Features with values appear as `"FeatureName: value"` in the output.

Click **Next: Settings** to continue.

Step 4 Configure Settings



What are Settings?

Settings control specific behaviors of your org — like enabling Lightning Experience, Chatter, encryption, password policies, etc.

What to do:

1. **Browse by category** — Click any category header to expand/collapse it
2. **Toggle ON/OFF** — Each setting has a toggle switch:
 - **Dark grey (off)** = not included in definition
 - **Purple/blue (on)** = will be enabled
3. **Use search** — Type in the search box to filter settings across all categories
4. **Check badge counts** — The counter (e.g., "2/3") shows enabled vs total settings per category

The "5 enabled" badge at the top shows total active settings across all categories.

Custom Settings

Data & Privacy 1/1 ▲

☒ **Enable Encryption Management**
Enables Shield Platform Encryption management.
encryptionManagementSettings.enableEncryptionManagement

Custom Settings + Add Row

For settings not listed above. Use dot-notation paths and values (true, false, or strings).

e.g. caseSettings.systemUserEmail	e.g. support@acme.com
-----------------------------------	-----------------------

Back Next: Review & Create

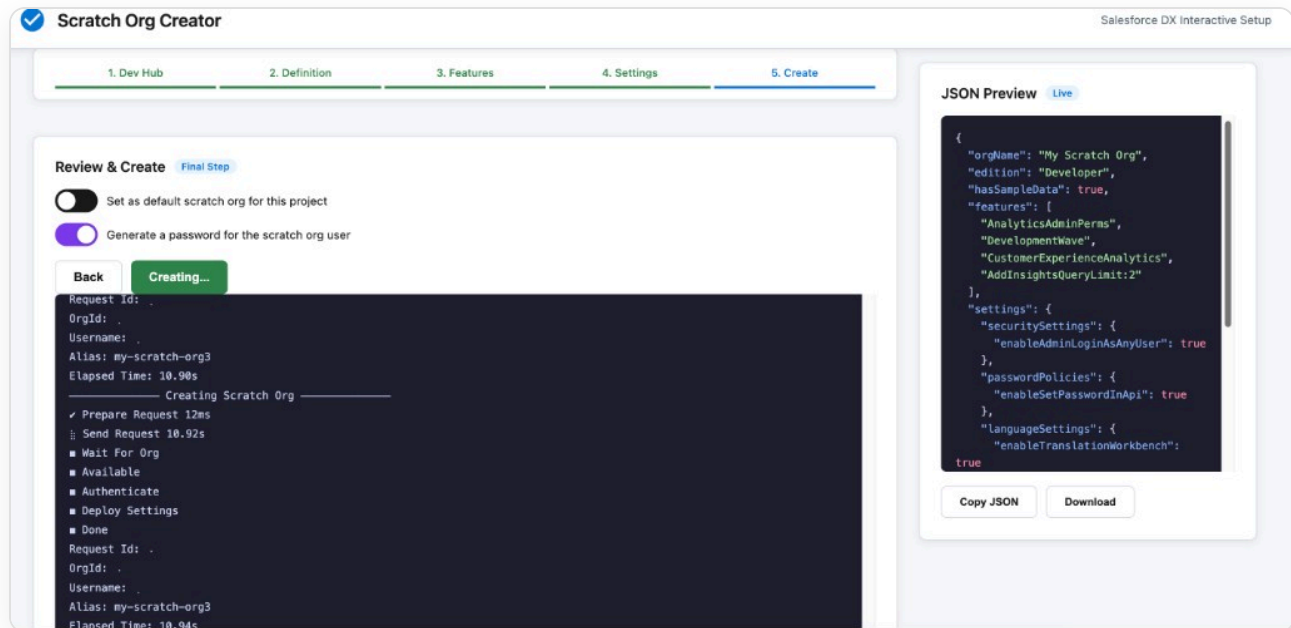
For settings not in the pre-built list:

1. Scroll to the "Custom Settings" section at the bottom
2. Click **+ Add Row**
3. Enter the **dot-notation path** in the left field (e.g., `caseSettings.systemUserEmail`)
4. Enter the **value** in the right field (e.g., `support@acme.com`)
5. Add additional rows as needed
6. Click × to remove any row

Note: Custom setting paths must follow Salesforce's scratch org definition format. Refer to [Salesforce documentation](#) for valid paths.

Click **Next: Review & Create** to continue.

Step 5 Review & Create



What to do:

1. **Review the JSON Preview** — The right panel shows the complete scratch org definition that will be used
2. **Set optional toggles:**
 - **Set as default scratch org** — ON if this should become your active working org
 - **Generate a password** — ON to auto-generate login credentials for the admin user
3. **Click "Create Scratch Org"**

During Creation:

The button changes to "Creating..." and you'll see real-time console output showing progress:

- **Prepare Request** — Building the org request

- **Send Request** — Sending to Salesforce (typically 10-15 seconds)
- **Wait For Org** — Salesforce is provisioning your org
- **Available** — Org has been created
- **Authenticate** — Setting up authentication
- **Deploy Settings** — Applying your selected settings
- **Done** — Org is ready to use!

After Success:

- A green card appears with your org credentials (username, password, instance URL)
 - Click **"Open Org in Browser"** to jump into the new org immediately
 - Use **"Copy JSON"** or **"Download"** buttons to save the definition file for reuse
-

Session Recovery

If you accidentally close the browser or the server stops mid-configuration:

- **Your progress is automatically saved** — All form data is stored locally
- **Just restart the script** — Run `bash scratch-org-creator.sh` again
- **Previous selections are restored** — A "Session Restored" banner appears
- You'll be returned to the exact step you were on with all data intact

To start completely fresh, click "Start Fresh" on the restore banner.

Troubleshooting

Problem	Solution
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"sf command not found"	Install Salesforce CLI: <code>npm install -g @salesforce/cli</code>
"python3 not found"	Install Python 3 from python.org
"Could not install Flask"	Run manually: <code>pip3 install flask</code>
Browser doesn't open	Copy the URL from terminal output and open manually
"Dev Hub is not enabled"	In your Salesforce org: Setup → Dev Hub → Toggle ON
Org creation fails	Check console output for error details — common causes: expired auth, invalid features
Port already in use	Kill existing servers: <code>kill \$(lsof -t -i:8484)</code>